A later partial obstruction for star-finite ball coverings

Literature-implied partial subcase for arXiv:2002.04308

Status

This is a `literature_implied_answer (partial strengthened subcase)` packet. It records a later theorem about ℓ_1 which is explicitly presented as a partial contribution to the star-finite covering problem in arXiv:2002.04308 [1].

Source question

In Section 3.4 of the source paper, the authors ask:

> Problem 3.21. Does there exist a separable Banach space admitting a star-finite covering by closed balls?

They also ask analogous questions for infinite-dimensional Frechet smooth spaces and for spaces $C(K)$.

Supporting theorem

The later paper *Tilings and coverings by balls in* ℓ_1, arXiv:2604.15092 [2], states Theorem A. Its second part says that, for every infinite cardinal $\kappa < 2^{\aleph_0}$ and every $n \in \mathbb{N}$, the space $\ell_1(\kappa)$ does not admit star-n-finite coverings by balls of positive radius. Taking $\kappa = \aleph_0$, this gives in particular:

$$\ell_1 \text{ has no star-}n\text{-finite covering by balls of positive radius} \quad \text{for every fixed } n.$$

The supporting paper explicitly describes this as a partial answer to the problem of existence of star-finite coverings by balls in ℓ_1.

Scope limitations

This does not settle Problem 3.21. A star-finite family only requires each ball to meet finitely many other balls, with no uniform bound on that finite number. A star-n-finite family is therefore a strictly stronger requirement. The full separable-Banach question, and even the exact star-finite question for ℓ_1, remain outside this packet.

References

[1] C. A. De Bernardi, J. Somaglia, and L. Vesely, *Star-finite coverings of Banach spaces*, arXiv:2002.04308.

[2] C. A. De Bernardi, T. Russo, S. Sezgek, and J. Somaglia, *Tilings and coverings by balls in* ℓ_1, arXiv:2604.15092.